Enumerative Combinatorics — problem 5.47

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We call a permutation $(x_1, \ldots, x_{2n})$ of the numbers $1, 2, \ldots, 2n$ pleasant if $|x_i - x_{i+1}| = n$ for at least one $i \in \{1, \ldots, 2n-1\}$. Prove that more than one-half of all permutations are pleasant for each positive integer n (IMO 1989).